



Discrimination, loneliness, and academic achievement among university students in Germany: results from the StudiBiFra study

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Abstract

Aim Discrimination is associated with poorer well-being and academic performance among university students. Loneliness during the study period may further undermine psychosocial well-being and academic development. This research examines how discrimination within the study environment, by fellow students or lecturers, is associated with loneliness, identification with university, and worsened academic achievement among university students in Germany. It also explores the role of loneliness as a moderator.

Subject and methods We analysed cross-sectional data from the Survey on Study Conditions and Mental Health of University Students (StudiBiFra) survey ($n = 24,533$), conducted across 13 German universities between 2021 and 2023. Perceived discrimination by lecturers and fellow students was assessed separately. Self-reported outcomes included loneliness, identification with university, and academic achievement. Multivariable logistic regression models were applied, adjusting for sociodemographic factors to account for the diversity of the sample. Moderation analyses assessed whether associations differed by loneliness.

Results Perceived discrimination within the study programme, by lecturers and fellow students, was significantly associated with higher odds of loneliness, no identification with university, and worsened academic achievement. Associations were generally stronger for discrimination by fellow students. Gender-diverse students showed higher odds, whereas students aged over 40 years had lower odds across outcomes. Loneliness did not moderate the relationship with academic achievement but moderated the association with identification with university, with weaker associations among lonely students.

Conclusion Perceived discrimination within higher education is closely linked to students' psychosocial well-being and academic outcomes. Our findings highlight the role of university environments in shaping inequalities and underscore the need for structural and institutional strategies that address discrimination and promote inclusive and supportive study conditions.

Keywords Discrimination · Loneliness · Identification with university · Academic achievement · Student belonging · Higher education

Laura Pilz González and Vanessa Wenig contributed equally to this work and share first authorship.

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Introduction

University students predominantly fall within a developmental stage characterised by significant social, academic, and structural transitions. As students navigate new and demanding environments such as universities, experiences of social inclusion and belonging play an essential role for their well-being (Arnett 2015; Murray and Arnett 2018). Conversely, experiences of social exclusion or marginalisation during this phase may be associated with adverse psychosocial and academic outcomes (Palmer et al. 2023). Central to this is exposure to discrimination, defined as the unequal or prejudicial treatment of individuals or groups on the basis of socially prescribed and constructed characteristics. Rather than merely a subjective experience, discrimination contributes to social inequalities across populations (Pascoe and Smart Richman 2009; Dovidio et al. 2010).

Within higher education, research consistently shows that students in marginalised social positions such as those related to race, ethnicity, sexual orientation, and/or gender often encounter discrimination (Hua et al. 2023; DeLaney et al. 2021; Qeadan et al. 2022; Stevens et al. 2018). Discrimination has been linked to a range of adverse academic outcomes including lower study satisfaction (Meyer et al. 2022) lower grades, or even higher dropout rates (Stevens et al. 2018; Mills 2021). It has also been associated with increased psychological distress, such as anxiety and depressive symptoms (Pilz González et al. 2025; Meyer et al. 2022).

While discrimination is often studied as a direct personal experience (Meyer et al. 2022), it may also be reflected in how students perceive their study environment (Pilz González et al. 2025). Even without direct targeting, observing discriminatory behaviour or the perception of discriminatory environments may foster feelings of insecurity and hypervigilance of potential discrimination (Pascoe and Smart Richman 2009; Emmer et al. 2024; Meyer et al. 2022). In this study, we focus on perceived discriminatory study environments within the university context, examining students' perceptions of discrimination from lecturers and fellow students in their study programme as indicators of the broader institutional climate.

As institutions where structural power dynamics are pronounced and deeply rooted, universities can exacerbate existing disadvantages and reinforce discrimination, particularly against already marginalised groups (Kaufmann 2016; Krell 2013). In Germany, where social mobility remains comparatively low, structural inequalities continue to shape access to and experiences within higher education (OECD 2018; Dippelhofer-Stiem 2017). Despite increasing student diversity, dominant representations of 'the typical student' continue to reflect privileged and traditional educational pathways, favouring students from high socio-economic backgrounds and/or of German origin (Kaufmann 2016; Kerst et al. 2024). Such structural inequalities may influence how students experience their study environment,

including exposure to exclusionary practices and unequal opportunities. From a public health perspective, these differences in social and institutional experiences may contribute to inequalities in psychosocial well-being and academic outcomes within higher education and beyond (Bowleg 2020).

Evidence on discrimination in German higher education reflects the extent of such inequalities. A nationwide survey revealed that approximately 26% of university students had experienced first-hand discrimination during their studies, primarily based on gender and non-German origin (Meyer et al. 2022). Complementing this, data from the cross-sectional Survey on Study Conditions and Mental Health of University Students (StudiBiFra) highlights that students frequently perceive discrimination within their study environment, particularly originating from lecturers (Pilz González et al. 2025). Together, these studies provide a more comprehensive picture of discrimination in German higher education by capturing both direct experiences of discrimination and the broader perception of discriminatory environments within institutional contexts.

Beyond academic achievement, perceived discrimination has also been associated with feelings of loneliness (Li Verdugo et al. 2024; Maleku et al. 2022) and reduced student identification with university (Mills 2021). Loneliness can be understood as a feeling of social disconnection arising from the discrepancy between desired and actual social relationships (Mansfield et al. 2021). Among university students, it has emerged as a pressing issue (Hysing et al. 2020; Wenig et al. 2023; Zahedi et al. 2022) frequently exacerbated by exclusionary practices and institutional bias within the academic settings (Li Verdugo et al. 2024; Maleku et al. 2022). Consequently, loneliness has been linked to reduced university connectedness and engagement (Wenig et al. 2025), as well as increased procrastination and underperformance (Jefferson et al. 2023; Freyhofer et al. 2021). At the same time, loneliness is not only a matter of social well-being, it is also associated with a range of negative mental health outcomes including depression, generalised anxiety, and suicidal ideation (Beutel et al. 2017; Leigh-Hunt et al. 2017; Hawkley 2022).

Similarly, students' identification with their university, reflecting their sense of belonging to their institution, is closely linked to psychological well-being and academic success (Pedler et al. 2022; Au et al. 2023; Pomyalova et al. 2020). A strong sense of belonging has been associated with lower dropout intentions and improved academic achievement (Moffitt et al. 2018; Trigueros et al. 2020; Pedler et al. 2022; Santini et al. 2021).

Despite growing research on discrimination and loneliness as independent risk factors in students' academic trajectories, little is known about how they relate to one another. In particular, loneliness may be associated with differences in how perceived discrimination relates to student outcomes and psychosocial well-being. Considering loneliness as an indicator

of social disconnection, it may reflect differences in how students experience and interpret their study environment.

Against this background, we hypothesise that perceived discrimination by fellow students or lecturers in their study environment is associated with loneliness, no identification with university, and poorer academic achievement. Furthermore, we explore whether these associations differ depending on whether or not students feel lonely. While exploring these associations, we also account for the sociodemographic diversity of our sample. Our aim is to provide insight into how perceptions of discrimination within the university environment are associated with students' well-being and academic performance. Ultimately, these insights could inform institutional strategies designed to support growing diverse student populations more effectively and promote more inclusive, supportive, and healthy study environments.

Methods

Data and sample

We used data from the cross-sectional StudiBiFra study, which assessed the study conditions and mental health of students at 13 German universities. Invitations to participate were sent via email to students aged 18 years and above enrolled in undergraduate and postgraduate programmes. Data collection was carried out using the LimeSurvey platform (LimeSurvey GmbH, Hamburg, Germany) over the period from June 2021 to March 2023, using the Bielefeld Questionnaire on Study Conditions and Mental Health (Burian et al. 2022) in both English and German. The average response rate was 11.4%, calculated on the basis of the number of enrolled students at the respective institutions. Further information about the study design and recruitment process is published elsewhere (Lehnchen et al. 2025; Heinrichs et al. 2024). Before starting the survey, all students gave their informed consent to participate. The study received approval from the Ethics Committee of Charité – University Medical Center Berlin (EA1/055/21).

Measures

All items were derived from the Bielefeld Questionnaire on Study Conditions and Mental Health (Burian et al. 2022), which has been used in previous analyses of the same data set on student mental health and study conditions (Wenig et al. 2025; Pilz González et al. 2025; Lehnchen et al. 2025; Heinrichs et al. 2024).

Independent variables

Perceived discrimination, the main predictor in our study, was assessed through two items. The first item measured discrimination by lecturers: *'My lecturers discriminate against individual students (e.g., on the basis of gender, disability, age, origin, racist attributions, pregnancy)'*. The second item assessed discrimination by fellow students: *'In my course of studies, individual students are discriminated against by other students (e.g., on the basis of gender, disability, age, origin, racist attributions, pregnancy)'*. Both items were measured as a five-point Likert-type scale ranging from 1 (*'strongly agree'*) to 5 (*'strongly disagree'*).

Dependent variables

Our analysis focused on the three dependent variables: loneliness, identification with university, and academic achievement. Each variable was assessed using a single-item measure with the same five-point Likert-type scale (loneliness: *'I am lonely in my studies'*; identification with university: *'I don't really care whether I am studying at [name of university] or another one'*; academic achievement: *'My academic achievement has deteriorated since the start of my studies'*). Academic achievement reflects students' subjective evaluation of changes in their academic achievement since the start of their studies.

All responses for the three variables were recoded into binary variables to allow comparison between those who strongly agreed or agreed (loneliness: yes; identification with university: no; academic achievement: worsened) and those who neither agreed nor disagreed, disagreed, or strongly disagreed (loneliness: no; identification with university: yes; academic achievement: maintained). This binary categorisation was chosen to facilitate interpretation.

Covariates

Due to the secondary nature of the data, the selection of variables to capture the diversity of our sample could not be determined from the outset. To guide our selection of covariates for this analyses, we employed the Diversity Minimal Item Set (DiMIS) as it aims to address the diversity and gender gap within health research. Grounded in anti-discrimination legislation established by the United Nations Human Rights Office, the DiMIS framework emphasises relevant domains that reflect the diversity of the sample beyond binary domains (Stadler et al. 2023). In addition, further analyses on the mental health of marginalised groups within this dataset, as reported by Heinrichs et al. (2024) highlighted disparities affecting specific subgroups. These findings underscored the relevance of capturing

sociodemographic dimensions of diversity in our own analysis. Consequently, four sociodemographic variables were selected to delineate the diversity of our sample, including gender, age, first academic generation status as a proxy for socio-economic status (Miethe et al. 2014; Dippelhofer-Stiem 2017), and family care tasks. Gender response options included male, female, and gender-diverse. Age was grouped into four categories: 18–25 years, 26–30 years, 31–40 years, and over 40 years. To assess first academic generation status, participants indicated whether they were the first in their family (parents/siblings) to attend university, with responses recorded as ‘Yes’ or ‘No’. Additionally, participants were asked about family care tasks, including care for children or taking care of relatives in need of care and/or supervision. For analysis purposes, responses were later categorised as ‘Yes’ (any caregiving tasks) or ‘No’ (no caregiving tasks). For all items analysed in this paper, students were given the option of choosing ‘no answer’.

Analyses

A series of six multivariable logistic regression models including multiple covariates were employed to examine a) the associations of perceived discrimination by fellow students with loneliness, identification with university, and academic achievement, and b) the associations of perceived discrimination by lecturers and the same three dependent variables. Perceived discrimination by lecturers and fellow

students were analysed separately to allow for clearer interpretation of their distinct associations with the outcomes. All models were adjusted for age, gender, first academic generation, and family care tasks. Since students’ experiences may vary across institutions (Pomyalova et al. 2020), we also included the institution as a dummy variable to account for potential differences.

Participants with missing data on the relevant variables for our analyses were excluded from the regression analyses. Multicollinearity was assessed among the independent variables in the models. Although correlations between the independent variables were low ($r < 0.70$), formal diagnostics were conducted using variance inflation factors (VIF) and tolerance values. VIF values ranged from 1.01 to 1.17, with corresponding tolerance values above 0.85, indicating no evidence of problematic multicollinearity. Logistic regression results were reported as odds ratios (ORs) with 95% confidence intervals (CIs).

Next, to examine whether associations of perceived discrimination by fellow students or lecturers with academic achievement and identification with university, respectively, differed by loneliness, interaction terms between perceived discrimination and loneliness were estimated using Hayes’ (2013) PROCESS macro, which is commonly applied to test moderation effects. The control variables were the same as in the multivariable regression models. Our hypothesised moderation paths and subsequent models 1–4 are depicted in Fig. 1. Moderation results were reported as regression

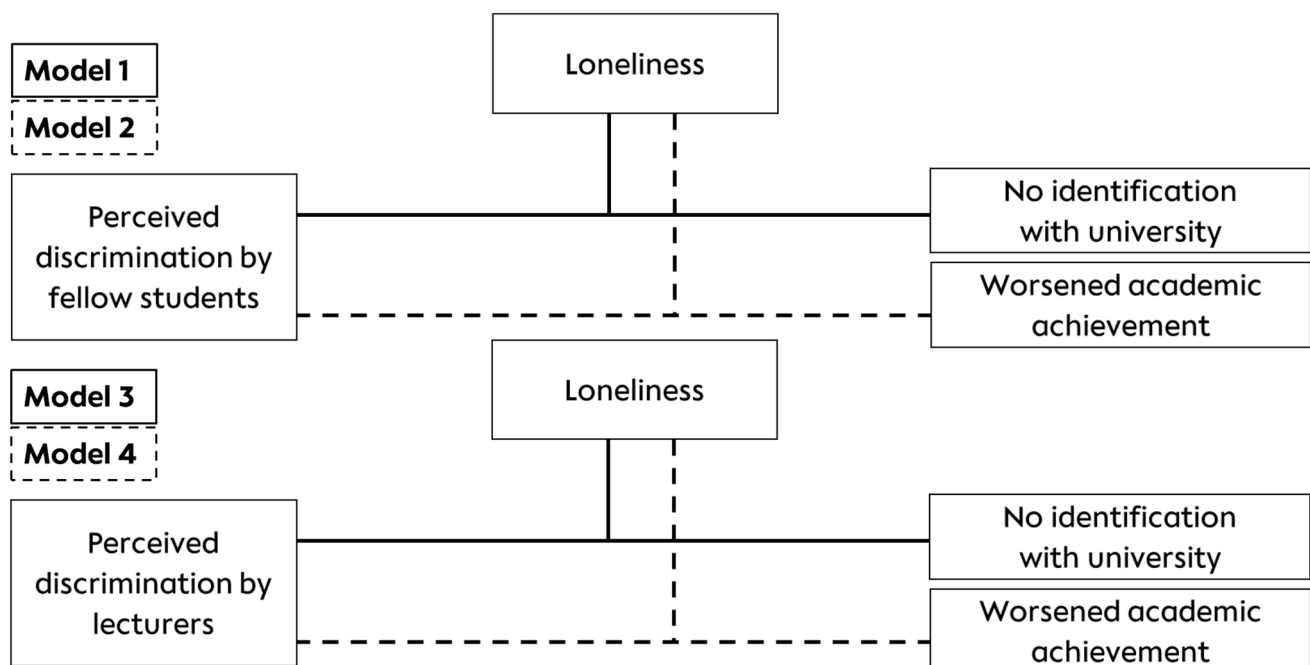


Fig. 1 Hypothesised paths for the interrelations of perceived discrimination by fellow students and lecturers with identification with the university, academic achievement, and the moderating role of loneliness

coefficients (log-odds) with 95% CIs. Statistical significance was assessed via two-tailed p -values ($p < 0.05$) for all models. Data analyses were carried out on IBM SPSS, version 27 with the PROCESS macros expansion.

Results

Sample characteristics

The sample consisted of 24,533 participating students, of whom the majority (66.9%) identified as female and

1.8% as gender-diverse. Most participants were aged 18–25 years (69.4%) and 37.9% identified as first-generation students. Additionally, 13.5% of participants reported family care tasks. Regarding the main variables, 4.8% of students perceived discrimination by fellow students and 6.1% by lecturers. Loneliness was reported by 33.8%, 31.0% showed no identification with university, and 31.9% of students reported a worsening of their academic performance since the start of their studies. The detailed descriptive statistics are provided in Table 1.

Factors associated with loneliness, identification with university, and academic achievement

The results for discrimination by fellow students as independent variable are detailed in Table 2. Students who strongly agreed they perceived discrimination by their fellow students had almost five times the odds of feeling lonely during their studies (OR = 4.69, 95% CI [3.25; 6.76]) compared with students who strongly disagreed. Likewise, they had 1.45 times the odds of showing no identification with university (95% CI [1.02; 2.06]) and almost three times the odds of reporting worsened academic achievement (OR = 2.92, 95% CI [1.93; 4.43]).

In addition, several sociodemographic variables were significantly associated with these outcomes, independently of perceived discrimination. Students identifying as gender-diverse had higher odds of reporting loneliness (OR = 1.56, 95% CI [1.14; 2.14]) and worsened academic achievement (OR = 2.02, 95% CI [1.36; 2.99]) compared with male students. Students aged over 40 years had lower odds of reporting loneliness (OR = 0.64, 95% CI [0.47; 0.87]), no identification with university (OR = 0.62, 95% CI [0.44; 0.87]), and worsened academic achievement (OR = 0.66, 95% CI [0.45; 0.98]) compared with students aged 18–25 years. First-generation students also had higher odds of loneliness (OR = 1.12, 95% CI [1.02; 1.21]) than those from academic families. Similarly, students with family care tasks also showed higher odds of worsened academic achievement (OR = 1.39, 95% CI [1.18; 1.64]) compared with students without such responsibilities.

Table 3 displays the results from perceived discrimination by lecturers as independent variable. Students who strongly agreed they perceived discrimination by lecturers had almost twice the odds for loneliness (OR = 1.92, 95% CI [1.41; 2.61]) and no identification with university (OR = 1.94, 95% CI [1.43; 2.64]) than students who did not. Additionally, they had 1.63 times the odds of reporting worsened academic achievement (95% CI [1.11; 2.38]).

As before, various sociodemographic factors showed significant associations with the outcomes, regardless of perceived discrimination by lecturers. Students identifying

Table 1 Sample characteristics ($n = 24,533$)

Variable	% (n)
Gender ($n = 14,592$)	
Male	31.26 (4562)
Female	66.92 (9765)
Diverse	1.82 (265)
Age in years ($n = 14,809$)	
18–25	69.42 (10,280)
26–30	19.22 (2847)
31–40	8.97 (1329)
> 40	2.38 (353)
First academic generation ($n = 14,667$)	
No	62.12 (9111)
Yes	37.88 (5556)
Family care tasks ($n = 13,426$)	
No	86.46 (11,608)
Yes	13.54 (1818)
Perceived discrimination by fellow students ($n = 13,185$)	
Strongly disagree	61.24 (8074)
Disagree	28.01 (3693)
Neither agree nor disagree	5.97 (787)
Agree	3.33 (439)
Strongly agree	1.46 (192)
Perceived discrimination by lecturers ($n = 14,304$)	
Strongly disagree	60.86 (8705)
Disagree	24.63 (3523)
Neither agree nor disagree	8.43 (1206)
Agree	4.20 (601)
Strongly agree	1.88 (269)
Loneliness ($n = 14,777$)	
No	66.19 (9781)
Yes	33.81 (4996)
Identification with university ($n = 14,152$)	
No	31.02 (4390)
Yes	68.98 (9762)
Academic achievement since start of studies ($n = 9752$)	
Maintained	68.09 (6640)
Worsened	31.91 (3112)

Table 2 Logistic regressions on the association of discrimination by fellow students with loneliness, identification with university, and academic achievement as dependent variables

Predictor variables	Loneliness (<i>n</i> = 11,483)		No identification with university (<i>n</i> = 11,106)		Worsened academic achievement since start of studies (<i>n</i> = 7417)	
	OR*	95% CI	OR*	95% CI	OR*	95% CI
Perceived discrimination by fellow students						
Strongly disagree	Ref.		Ref.		Ref.	
Disagree	1.12	1.02; 1.24	1.09	0.99; 1.20	1.15	1.02; 1.30
Neither agree nor disagree	1.90	1.59; 2.26	1.14	0.95; 1.37	1.30	1.04; 1.63
Agree	3.09	2.46; 3.89	1.65	1.31; 2.08	1.46	1.08; 1.97
Strongly agree	4.69	3.25; 6.76	1.44	1.02; 2.06	2.92	1.93; 4.43
Covariates	OR*	95% CI	OR*	95% CI	OR*	95% CI
Gender						
Male	Ref.		Ref.		Ref.	
Female	1.11	1.01; 1.22	1.02	0.93; 1.12	1.07	0.95; 1.20
Diverse	1.56	1.14; 2.14	1.33	0.97; 1.83	2.02	1.36; 2.99
Age						
18–25	Ref.		Ref.		Ref.	
26–30	1.07	0.96; 1.20	1.07	0.96; 1.19	1.10	0.96; 1.25
31–40	0.81	0.69; 0.96	0.91	0.77; 1.07	0.89	0.73; 1.08
> 40	0.64	0.47; 0.87	0.62	0.44; 0.87	0.66	0.45; 0.98
First academic generation						
No	Ref.		Ref.		Ref.	
Yes	1.12	1.02; 1.21	0.97	0.89; 1.02	0.98	0.88; 1.10
Family care tasks						
No	Ref.		Ref.		Ref.	
Yes	1.04	0.91; 1.19	0.89	0.77; 1.02	1.39	1.18; 1.64

Notes: OR = odds ratio; CI = confidence interval; Ref. = reference category; *ORs adjusted for all other variables in the table as well as university institution (not reported). Estimates in bold are statistically significant ($p < 0.05$)

as gender-diverse had higher odds for all three outcomes than male students. Those over 40 years showed lower odds for loneliness (OR = 0.63, 95% CI [0.47; 0.84]), no identification with university (OR = 0.57, 95% CI [0.41; 0.79]), and worsened academic outcomes (OR = 0.63, 95% CI [0.43; 0.91]) than students aged 18–25 years. First-generation students also reported higher odds of loneliness (OR = 1.14, 95% CI [1.05; 1.24]) than those from academic families. Again, students with family care tasks showed higher odds of worsened academic achievement (OR = 1.44, 95% CI [1.23; 1.69]) than students without such responsibilities.

Differences in strengths of associations by loneliness

When looking at perceived discrimination by fellow students and identification with university (model 1), the interaction term was statistically significant (Coeff = −0.10, 95% CI [−0.20; −0.01]), suggesting that the association between perceived discrimination by fellow students and no identification

with university differed by loneliness. The negative interaction term suggests that the association was weaker for lonely students. In contrast, for worsened academic achievement (model 2), the interaction term was not statistically significant (Coeff = 0.00, 95% CI [−0.12; 0.12]), indicating no evidence that the association between perceived discrimination by fellow students and academic achievement differed by loneliness.

Concerning the association between perceived discrimination by lecturers and no identification with university (model 3), the interaction term was statistically significant (Coeff = −0.10, 95% CI [−0.19; −0.02]), suggesting that the association between perceived discrimination by lecturers and no identification with university differed (i.e., attenuated) by loneliness. Again, the negative interaction term suggests that the association was weaker for lonely students compared with those who are not lonely. For worsened academic achievement (model 4), the interaction term was not significant (Coeff = −0.01, 95% CI [−0.11; 0.10]), indicating no difference by loneliness.

Table 3 Logistic regressions on the association of discrimination by lecturers with loneliness, identification with university, and academic achievement as dependent variables

Predictor variables	Loneliness (<i>n</i> = 11,483)		No identification with university (<i>n</i> = 11,106)		Worsened academic achievement since start of studies (<i>n</i> = 7417)	
	OR*	95% CI	OR*	95% CI	OR*	95% CI
Perceived discrimination by lecturers						
Strongly disagree	Ref.		Ref.		Ref.	
Disagree	1.19	1.08; 1.30	1.15	1.05; 1.27	1.23	1.09; 1.39
Neither agree nor disagree	1.50	1.29; 1.74	1.47	1.27; 1.71	1.21	1.01; 1.47
Agree	1.67	1.36; 2.04	1.74	1.42; 2.12	1.73	1.35; 2.22
Strongly agree	1.92	1.41; 2.61	1.94	1.43; 2.64	1.63	1.11; 2.38
Covariates	OR*	95% CI	OR*	95% CI	OR*	95% CI
Gender						
Male	Ref.		Ref.		Ref.	
Female	1.13	1.03; 1.23	1.04	0.96; 1.14	1.09	0.97; 1.22
Diverse	1.91	1.42; 2.58	1.44	1.07; 1.94	2.00	1.39; 2.89
Age						
18–25	Ref.		Ref.		Ref.	
26–30	1.16	1.04; 1.28	1.07	0.96; 1.18	1.15	1.01; 1.31
31–40	0.86	0.74; 1.00	0.93	0.79; 1.08	0.91	0.76; 1.10
> 40	0.63	0.47; 0.84	0.57	0.41; 0.79	0.63	0.43; 0.91
First academic generation						
No	Ref.		Ref.		Ref.	
Yes	1.14	1.05; 1.24	0.97	0.89; 1.05	1.03	0.93; 1.15
Family care tasks						
No	Ref.		Ref.		Ref.	
Yes	1.08	0.95; 1.23	0.90	0.79; 1.03	1.44	1.23; 1.69

Notes: OR = odds ratio; CI = confidence interval; Ref. = reference category; *ORs adjusted for all other variables in the table as well as university institution (not reported). Estimates in bold are statistically significant ($p < 0.05$)

Moderation plots for both the statistically significant and non-significant results can be found in our supplementary file.

Discussion

Our study examined how perceptions of discrimination within the university environment are associated with loneliness, identification with university, and academic achievement among university students in Germany. We also explored whether the associations between perceived discrimination and identification with university, as well as academic achievement, differed depending on loneliness.

Consistent with our hypotheses, our results indicate a strong association between perceived discrimination in the study environment, whether from lecturers or fellow students, and all three outcomes. These findings align with prior research linking discrimination to adverse psychosocial (Li Verdugo et al. 2024; Maleku et al. 2022; Schmitt et al. 2014) and academic outcomes (Jackson and Reynolds 2013; Brittian et al. 2015; Pomyalova et al. 2020). Building

on this body of work, our study focuses on students' perceptions of discrimination within their learning environment and suggests that perceived institutional climates are associated with students' well-being and academic functioning. Consequently, our findings demonstrate that discrimination operates at the interpersonal level and as a social determinant within educational settings, contributing to inequalities in mental health and academic outcomes.

The strongest associations were observed between perceived discrimination and loneliness. Approximately one in three students reported feeling lonely during their studies. This high prevalence further emphasises the relevance of loneliness as a public health concern within higher education (Hysing et al. 2020; Wenig et al. 2023; Zahedi et al. 2022). The stronger associations observed for perceived discrimination by fellow students, compared to lecturers, with loneliness may reflect the central role of peer relationships in students' everyday social environments (Maunder 2018). Perceived discrimination among peers may directly disrupt opportunities for social integration, thereby increasing feelings of loneliness (Goodfellow et al. 2022). By contrast,

the relationship with lecturers may play a different role in a university setting than e.g. the student–teacher relationship in school contexts. Interactions with lecturers in higher education are typically less frequent and more academically focused than student–teacher relationships in school settings, where teachers often play a more central role in students’ daily social and emotional lives (Hagenauer and Voles 2014). Consequently, discrimination attributed to lecturers, while still significant, may operate more indirectly through perceptions of institutional norms and fairness (Li Verdugo et al. 2024).

A similar pattern emerged for academic achievement, with stronger associations observed with perceived discrimination by fellow students than by lecturers. This finding further indicates that peer-related experiences are more closely tied to students’ day-to-day academic engagement, including collaboration, participation, and access to informal support networks (Li Verdugo et al. 2024). However, when considering identification with university, stronger associations were observed for perceived discrimination by lecturers. This finding likely reflects how lecturers are seen as representatives of the institution. Therefore, perceived discrimination from lecturers in the study environment may more directly shape students’ sense of belonging and identification with the university as a whole (Kohútová and Kohút 2024; Fenaughty et al. 2019).

Independent from perceived discrimination, several sociodemographic characteristics were significantly associated with the three outcomes. Students identifying as gender-diverse had higher odds of loneliness, worsened academic achievement, and no identification with university. First-generation students reported higher odds of loneliness, while students with family care tasks were more likely to report worsened academic achievement. In contrast, older students (≥ 40 years) showed lower odds across all outcomes. These patterns suggest that psychosocial risks are unequally distributed across student groups and reflect a complex interplay between social position and structural conditions within higher education (Wilson and Liss 2023; Hajek et al. 2023), thus emphasising the need for targeted and equity-oriented interventions.

The results of our moderation analysis yielded nuanced insights. Contrary to our initial expectations, loneliness did not moderate the association between perceived discrimination—whether from lecturers or fellow students—and worsened academic achievement. In other words, the association between perceived discrimination and worsened academic achievement did not differ according to students’ feelings of loneliness. This suggests that perceived discrimination and loneliness may be associated with academic outcomes independently of each other.

In contrast, loneliness emerged as a significant moderator in the relationship between perceived discrimination from

both lecturers and fellow students and no identification with university. Interestingly, the moderation effects diverge from our hypothesis: the association between discrimination and no identification with university was weaker for students experiencing loneliness than for their non-lonely fellow students. This suggests a more complex relationship than initially anticipated. One possible interpretation for this unexpected moderation effect is that lonely students may already feel disconnected from their university (Wenig et al. 2025), resulting in lower expectations of belonging. This, in turn, may render perceived discrimination a less influential component of their sense of identification with university. However, this interpretation should be considered in light of our operationalisation of identification with university, which was based on a single-item measure and may also capture broader attitudes towards the institution rather than identification alone. So, while these interpretations remain speculative due to the cross-sectional design of our study, we suggest that perceived discrimination and loneliness may operate through partially overlapping pathways in shaping students’ sense of belonging.

Moreover, the significant moderation effect of loneliness on identification with university underscores the need for universities to consider how institutional environments engage students who are already socially disconnected. If loneliness is associated with lower levels of identification with university irrespective of perceived discrimination, conventional approaches to fostering inclusion may not adequately reach these students (Mauder 2018). This suggests that efforts to strengthen students’ sense of belonging should not only address discriminatory experiences but also more broadly target social disconnection within the university context.

Taken together, our findings highlight perceived discrimination and loneliness as interrelated aspects of students’ study environments that are associated with both psychosocial well-being and academic outcomes. As such, our results position universities as critical settings for health promotion and reduction of social inequalities among young adults. Perceived discrimination emerges as a social determinant of health within this context, contributing to both psychosocial distress and unequal academic participation and performance. Addressing these challenges therefore requires moving beyond individual-level approaches towards structural and institutional strategies, including the implementation of anti-discrimination policies, inclusive teaching practices, and accessible mechanisms for reporting and addressing discriminatory behaviour. At the same time, initiatives that foster social connectedness, particularly for socially disadvantaged student groups, may help mitigate loneliness and strengthen institutional identification, although reducing loneliness alone is unlikely to be sufficient to counteract the effects of discrimination.

The complexity of these interactions highlights the need for further research to better understand the underlying mechanisms at play, linking perceived discrimination, loneliness, and academic outcomes in higher education. In particular, future studies should examine how different forms of social inequality and institutional contexts shape students' experiences and well-being over time.

Strengths and limitations

Our study has notable strengths. Both the multi-site nature of our data and large sample size enhance the robustness of our findings. Additionally, by controlling for sociodemographic variables representing key diversity dimensions from the DiMIS framework (Stadler et al. 2023), such as gender, caregiving tasks, first-generation status, and age, we offer a nuanced understanding of sociodemographic diversity that extends beyond traditional binary categorisations. A particularly notable strength of our study is the inclusion of gender-diverse students in the analyses of risk factors for loneliness, identification with university, and academic achievement—a contribution that is especially significant given that no similar studies have previously examined this population within this framework.

While our study provides valuable insights, several limitations must also be acknowledged. First, the low response rate in combination with the convenience sampling method may have introduced a self-selection bias, with vulnerable subgroups being potentially over- or underrepresented, as both lower engagement and greater motivation to report experiences of discrimination are plausible. Future research could compare subgroup representation across samples to assess the extent of any systematic bias. Although the large and multi-site sample may mitigate some concerns, the potential influence of selection bias cannot be ruled out and may limit the generalisability of the findings.

Second, as our analyses are based on a generic student health and well-being survey, we relied on self-reported measures of perceived discrimination rather than existing, validated measures of discrimination. It is also critical to highlight that the survey did not ask about the student's own experience with discrimination, but about whether students in general were discriminated against by fellow students or lecturers in their particular study programme. Consequently, our results reflect a broader perception of discrimination within the study environment rather than specific personal experiences. While this may have led to an over- or underestimation of discrimination, it also captures the perceived institutional climate, which is relevant to student well-being in itself. Future research should distinguish between different and multiple forms of discrimination including validated measures (Scheim and Bauer 2019) to enable a

more nuanced exploration, incorporating an intersectional perspective on how various identities interact to shape students' experiences (Kaufmann 2016). Qualitative research could further deepen these insights.

Furthermore, all key constructs, including perceived discrimination and loneliness were assessed using single-item measures, which may limit reliability and the ability to capture the full complexity of these constructs. However, single-item measures are commonly used in research on student populations within university contexts (Gestsdottir et al. 2021; Hysing et al. 2020) and may be particularly suitable for capturing context-specific experiences with minimal respondent burden. Moreover, as no validated measure of students' loneliness (Qualter et al. 2025) currently exists, a context-specific single-item measurement may offer advantages over broader scales that do not aim to capture the dimensions of loneliness experienced by young people. Nevertheless, our focus on university-specific loneliness may have led to underreporting of general loneliness (Shiovitz-Ezra and Ayalon 2012), which may also be important for students' well-being and requires further research.

In the case of identification with university, the single-item measure may not fully capture the multidimensional nature of this construct. Agreement with the statement may reflect factors other than identification with the own university, for example perceiving different universities as equally suitable instead of lacking identification with the own institution. Consequently, our findings relating identification with university should be interpreted in light of this specific operationalisation. Similarly, academic achievement since start of studies was assessed using a subjective self-reported measure rather than objective indicators such as examination grades or grade point average. Consequently, students' perceptions of changes in their academic achievement may not fully correspond to their actual academic performance.

An additional limitation relates to the recoding of all three dependent variables into binary categories, which may have reduced variability and obscured more nuanced differences in students' experiences. In particular, respondents who neither disagreed nor strongly disagreed were grouped together with those who disagreed or strongly disagreed. As neutral responses may reflect heterogeneous attitudes, some degree of misclassification cannot be ruled out. However, the binary categorisation facilitated interpretability and comparability across models.

Moreover, data collection partially occurred during restrictive COVID-19 measures in Germany, especially in the first year of data assessment (spring 2021 to spring 2022). Given the social distancing requirements and online teaching environment, research so far has shown that the influence of the COVID-19 restrictive measures acted as substantial stressors, potentially influencing the generalisability of our findings. To

assess the potential influence of the COVID-19 pandemic on our findings, we conducted a sensitivity analysis by repeating the main logistic regression models separately for data collected during periods of strict COVID-19 restrictions and after the restrictions had been relaxed. The overall pattern and magnitude of the associations remained comparable before and after the COVID-19 restrictions. Furthermore, it is important to note that research conducted before and after the COVID-19 pandemic consistently highlighted significant levels of loneliness among university students (Wenig et al. 2025; Haikalis et al. 2021), suggesting that this issue extends beyond the unique circumstances of COVID-19.

Finally, it is crucial to acknowledge that the relationships between discrimination, university outcomes, and student well-being are likely to be complex and cannot be assumed to operate in a single direction. For instance, research on older adults has shown that perceived discrimination can lead to social exclusion, which in turn heightens feelings of loneliness and marginalisation (Lee and Bierman 2019; Huxhold et al. 2022). Given the cross-sectional design of our study, the directionality of these associations cannot be determined. Further longitudinal studies are needed to better understand the relationships between discrimination, loneliness, and academic outcomes.

Conclusion

Our study contributes to the understanding of how perceived discrimination within the study environment is associated with loneliness, identification with university, and academic achievement among university students in Germany. The findings show that perceived discriminatory environments are significantly associated with feelings of loneliness, no identification with university, and worsened academic performance. Particularly socially disadvantaged groups, such as gender-diverse students, first-generation students, and those with caregiving tasks, face heightened risks, emphasising the need for targeted and equity-oriented support mechanisms within higher education.

By actively engaging with diverse student populations and addressing the structural causes of discrimination, universities can strengthen students' sense of belonging and institutional identification and thereby contribute to improved student mental health and academic outcomes. These findings underline the role of universities as key environments in shaping students' well-being and in addressing inequalities in academic outcomes. To achieve this, universities should seek to better understand the root causes of discrimination, implement appropriate structural and educational interventions, and raise awareness of its links to student well-being and academic outcomes. Participatory approaches involving university

leadership, teaching staff, students, and support services are essential for developing inclusive institutional environments.

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Data availability The data are available upon request from the authors.

Code availability Not applicable.

Declarations

Ethical considerations This study was performed in line with the principles of the Declaration of Helsinki. Approval was granted by the Ethics Committee of Charité – University Medical Center Berlin (Date: 26.03.2021/No. EA1/055/21).

Consent to participate Informed consent was obtained from all individual participants included in the study.

Conflict of interest The authors have no conflict of interest to declare that is relevant to the content of this article.

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References

- Arnett JJ (2015) College students as emerging adults: the developmental implications of the college context. *Emerg Adulthood* 4:219–222. <https://doi.org/10.1177/2167696815587422>

- Au A, Caltabiano NJ, Vaksman O (2023) The impact of sense of belonging, resilience, time management skills and academic performance on psychological well-being among university students. *Cogent Educ* 10:2215594. <https://doi.org/10.1080/2331186X.2023.2215594>
- Beutel ME, Klein EM, Brähler E, Reiner I, Jünger C, Michal M, Wiltink J, Wild PS, Münzel T, Lackner KJ, Tibubos AN (2017) Loneliness in the general population: prevalence, determinants and relations to mental health. *BMC Psychiatry* 17:97. <https://doi.org/10.1186/s12888-017-1262-x>
- Bowleg L (2020) Evolving intersectionality within public health: from analysis to action. *Am J Public Health* 111:88–90. <https://doi.org/10.2105/AJPH.2020.306031>
- Brittian AS, Kim SY, Armenta BE, Lee RM, Umaña-Taylor AJ, Schwartz SJ, Villalta IK, Zamboanga BL, Weisskirch RS, Juang LP, Castillo LG, Hudson ML (2015) Do dimensions of ethnic identity mediate the association between perceived ethnic group discrimination and depressive symptoms? *Cult Divers Ethnic Minor Psychol* 21:41–53. <https://doi.org/10.1037/a0037531>
- Burian J, Lehnchen J, Heumann E, Helmer SM, Stock C (2022) Bielefeld questionnaire on study conditions and health as a basis for the design of SGM at universities. In: Timmann M, Paack T, Fischer J, Steinke B, Dold C, Preuß M, Sprenger M (eds) *Handbook on student health management – perspectives, ideas and practical insights*. Springer, pp 77–87
- DeLaney EN, Williams CD, Elias MJ, Walker CJ, Smith TH, Adkins A, Lozada FT, Dick DM (2021) Racial discrimination and depressive symptoms mediated by conversations about race among students of color. *J Am Coll Health* 71:2835–2839. <https://doi.org/10.1080/07448481.2021.1998071>
- Dippelhofer-Stiem B (2017) Not entirely without. Disadvantage of working-class students in higher education. [Nicht ganz ohne. Benachteiligung von Arbeiterkindern im Studium]. *Die Hochschule: Journal für Wissenschaft und Bildung* 26:129–141. <https://doi.org/10.25656/01:16655>
- Dovidio JF, Hewstone M, Glick P, Esses VM (2010) Prejudice, stereotyping and discrimination: theoretical and empirical overview. In: Dovidio JF, Hewstone M, Glick P, Esses VM (eds) *The sage handbook of prejudice, stereotyping and discrimination*. SAGE Publications Ltd, pp 3–28
- Emmer C, Dorn J, Mata J (2024) The immediate effect of discrimination on mental health: a meta-analytic review of the causal evidence. *Psychol Bull* 150:215–252. <https://doi.org/10.1037/bul0000419>
- Fenaughty J, Lucassen MFG, Clark T, Denny S (2019) Factors associated with academic achievement for sexual and gender minority and heterosexual cisgender students: Implications from a nationally representative study. *J Youth Adolesc* 48:1883–1898. <https://doi.org/10.1007/s10964-019-01124-w>
- Freyhofer S, Ziegler N, de Jong E M and Schippers M C (2021) Depression and anxiety in times of COVID-19: How coping strategies and loneliness relate to mental health outcomes and academic performance. *Front Psychol* 12. <https://doi.org/10.3389/fpsyg.2021.682684>
- Gestsdóttir S, Gísladóttir T, Stefánsdóttir R, Johannsson E, Jakobsdóttir G, Rognvaldsdóttir V (2021) Health and well-being of university students before and during COVID-19 pandemic: a gender comparison. *PLoS ONE*. <https://doi.org/10.1371/journal.pone.0261346>
- Goodfellow C, Hardoon D, Inchley J, Leyland AH, Qualter P, Simpson SA, Long E (2022) Loneliness and personal well-being in young people: moderating effects of individual, interpersonal, and community factors. *J Adolesc* 94:554–568. <https://doi.org/10.1002/jad.12046>
- Hagenauer G, Voles SE (2014) Teacher-student relationship at university: an important yet under-researched field. *Oxf Rev Educ* 40:370–388. <https://doi.org/10.1371/10.1080/03054985.2014.921613>
- Haikalas M, Doucette H, Meisel MK, Birch K, Barnett NP (2021) Changes in college student anxiety and depression from pre- to during-COVID-19: Perceived stress, academic challenges, loneliness, and positive perceptions. *Emerg Adulthood* 10:534–545. <https://doi.org/10.1177/21676968211058516>
- Hajek A, König H-H, Blessmann M, Grupp K (2023) Loneliness and social isolation among transgender and gender diverse people. *Healthcare* 11:1517. <https://doi.org/10.3390/healthcare11101517>
- Hawkey LC (2022) Loneliness and health. *Nat Rev Dis Primers* 8:22. <https://doi.org/10.1038/s41572-022-00355-9>
- Hayes AF (2013) Introduction to mediation, moderation, and conditional process analysis: a regression-based approach. Guilford Press
- Heinrichs K, Lehnchen J, Burian J, Deptolla Z, Heumann E, Helmer S, Kucenko S, Stock C (2024) Mental and physical well-being among students in Germany: Results from the studibifra study. *J Public Health* 34:1119–1130. <https://doi.org/10.1007/s10389-024-02348-2>
- Hua J, Johnson AE, Hussain M, Howell JL (2023) Associations between perceived everyday discrimination and health among college students at a Hispanic-serving institution. *J Health Psychol* 28:648–662. <https://doi.org/10.1177/13591053221083739>
- Huxhold O, Suanet B, Wetzel M (2022) Perceived social exclusion and loneliness: two distinct but related phenomena. *Sociol Sci* 9:430–453. <https://doi.org/10.15195/v9.a17>
- Hysing M, Petrie KJ, Bøe T, Lønning KJ, Sivertsen B (2020) Only the lonely: a study of loneliness among university students in Norway. *Clin Psychol Eur*. <https://doi.org/10.32872/cpe.v2i1.2781>
- Jackson BA, Reynolds JR (2013) The price of opportunity: race, student loan debt, and college achievement. *Sociol Inq* 83:335–368. <https://doi.org/10.1111/soin.12012>
- Jefferson R, Barreto M, Jones F, Conway J, Chohan A, Madsen KR, Verity L, Petersen KJ, Qualter P (2023) Adolescent loneliness across the world and its relation to school climate, national culture and academic performance. *Br J Educ Psychol* 93:997–1016. <https://doi.org/10.1111/bjep.12616>
- Kaufmann ME (2016) Diversity not without intersectionality. [Diversity nicht ohne Intersektionalität]. In: Genkova P, Ringeisen T (eds) *Handbook on Diversity Competence: Volume 1: Perspectives and Areas of Application [Handbuch Diversity Kompetenz: Band 1: Perspektiven und Anwendungsfelder]*. Springer Fachmedien Wiesbaden, pp 819–837
- Kerst C, Kroher M, Steinkühler J (2024) Heterogeneous student body – on the social structure and study situation of students in Germany. [Heterogene Studierendenschaft – Zur Sozialstruktur und Studiensituation der Studierenden in Deutschland]. *Diskurs Kindheits- und Jugendforschung* 18:521–543. <https://doi.org/10.3224/diskurs.v18i4.06>
- Kohútová V, Kohút M (2024) The role of age, university attendance and other demographic factors in experiencing emerging adulthood. *Emerg Adulthood* 12:264–270. <https://doi.org/10.1177/21676968231226253>
- Krell G (2013) Multiple perspectives on diversity: Explore, reveal, create. [Vielfältige Perspektiven auf Diversity: erkunden, enthüllen, erzeugen]. In: Bender S-F, Schmidbaur M, Wolde A (eds) *Un-covering diversity: the scope and limits of diversity policies in higher education [Diversity ent-decken: Reichweiten und Grenzen von Diversity Policies an Hochschulen]*. Beltz Juventa, pp 61–79
- Lee Y, Bierman A (2019) Loneliness as a mediator of perceived discrimination and depression: examining education contingencies.

- Int J Aging Hum Dev 89:206–227. <https://doi.org/10.1177/0091415018763402>
- Lehnchen J, Helmer SM, Heinrichs K, Burian J, Deptolla Z, Heumann E, Stock C (2025) Assessment of study conditions, needs for action and students' mental health in Germany – results from the cross-sectional StudiBiFra study. *J Public Health (Oxf)*. <https://doi.org/10.1007/s10389-024-02387-9>
- Leigh-Hunt N, Baggeley D, Bash K, Turner V, Turnbull S, Valtorta N, Caan W (2017) An overview of systematic reviews on the public health consequences of social isolation and loneliness. *Public Health (Oxf)* 152:157–171. <https://doi.org/10.1016/j.puhe.2017.07.035>
- Li Verdugo J, Kong Z, Sembukutti Liyanage DS, Keum BT, Moody MD, Oh HY (2024) Associations between vicarious discrimination and mental health among young adult college students: findings from the 2020–2021 Healthy Minds Study. *J Affect Disord* 361:760–767. <https://doi.org/10.1016/j.jad.2024.06.082>
- Maleku A, Kim YK, Kirsch J, Um MY, Haran H, Yu M, Moon SS (2022) The hidden minority: Discrimination and mental health among international students in the US during the COVID-19 pandemic. *Health Soc Care Commun* 30:e2419–e2432. <https://doi.org/10.1111/hsc.13683>
- Mansfield L, Victor C, Meads C, Daykin N, Tomlinson A, Lane J, Gray K and Golding A (2021) A conceptual review of loneliness in adults: Qualitative evidence synthesis. *Int J Environ Res Public Health* 18. <https://doi.org/10.3390/ijerph18211522>
- Mauder RE (2018) Students' peer relationships and their contribution to university adjustment: the need to belong in the university community. *J Further High Educ* 42:756–768. <https://doi.org/10.1080/0309877X.2017.1311996>
- Meyer J, Strauß S and Hinz T (2022) The student survey in Germany: Focus analyses on experiences of discrimination at universities. [Die Studierendenbefragung in Deutschland: Fokusanalysen zu Diskriminierungserfahrungen an Hochschulen]. Hannover, Deutsches Zentrum für Hochschul- und Wissenschaftsforschung. https://doi.org/10.34878/2022.06.dzhw_brief
- Miethe I, Boysen W, Grabowsky S and Kludt R (2014) The situation as a first generation student at the university. [Die Situation als first generation student an der Hochschule]. First-generation students at German universities – self-organisation and the study experience, as illustrated by the 'Arbeiter Kind Forschung' initiative run by the Hans Böckler Foundation [First generation students an deutschen Hochschulen - Selbstorganisation und Studiensituation am Beispiel der Initiative Arbeiter Kind Forschung aus der Hans-Böckler-Stiftung]. 1 ed. Nomos Verlagsgesellschaft mbH & Co., pp 89–154
- Mills KJ (2021) Black students' perceptions of campus climates and the effect on academic resilience. *J Black Psychol* 47:354–383. <https://doi.org/10.1177/00957984211001195>
- Moffitt U, Juang LP, Syed M (2018) "We don't do that in Germany!" A critical race theory examination of Turkish heritage young adults' school experiences. *Ethnicities* 19:830–857. <https://doi.org/10.1177/1468796818788596>
- Murray JL, Arnett JJ (2018) Emerging adulthood and higher education: a new student development paradigm. Routledge
- OECD (2018) A broken social elevator? How to promote social mobility. Organisation for Economic Co-operation and Development Publishing. <https://doi.org/10.1787/9789264301085-en>
- Palmer AN, Jung E, Cobb RJ, Patel M (2023) It's complicated: everyday discrimination across the transition into adulthood. *Emerg Adulthood* 11:669–683. <https://doi.org/10.1177/21676968231152515>
- Pascoe EA, Smart Richman L (2009) Perceived discrimination and health: a meta-analytic review. *Psychol Bull* 135:531–554. <https://doi.org/10.1037/a0016059>
- Pedler ML, Willis R, Nieuwoudt JE (2022) A sense of belonging at university: student retention, motivation and enjoyment. *J Further High Educ* 46:397–408. <https://doi.org/10.1080/0309877X.2021.1955844>
- Pilz González L, Alonso-Perez E, Lehnchen J, Deptolla Z, Heumann E, Tezcan-Güntekin H, Heinrichs K, Stock C (2025) Mental health and the intersection of perceived discrimination and social inequalities among students in Germany - a quantitative intersectional study. *Int J Public Health* 69:1607826. <https://doi.org/10.3389/ijph.2024.1607826>
- Pomyalova VO, Volkova NV, Kalinina OV (2020) Effect of the university organizational culture perception on students' commitment: the role of organizational identification. *IOP Conf Ser Mater Sci Eng*. <https://doi.org/10.1088/1757-899X/940/1/012099>
- Qeadan F, Madden EF, Barbeau WA, Mensah NA, Azagba S, English K (2022) Associations between discrimination and adverse mental health symptoms and disorder diagnoses among college students in the United States. *J Affect Disord* 310:249–257. <https://doi.org/10.1016/j.jad.2022.05.026>
- Qualter P, Verity L, Walibhai W, Fuhrmann D, Riddleston L, Alam I, Conway J, Lau JYF (2025) Exploring child and youth understanding of loneliness through qualitative insights and evaluating loneliness measures considering those lived experiences. *Ann N Y Acad Sci* 1544:42–54. <https://doi.org/10.1111/nyas.15269>
- Santini ZI, Pisinger VSC, Nielsen L, Madsen KR, Nelausen MK, Koyanagi A, Koushede V, Roffey S, Thygesen LC, Meilstrup C (2021) Social disconnectedness, loneliness, and mental health among adolescents in Danish high schools: a nationwide cross-sectional study. *Front Behav Neurosci* 15:632906. <https://doi.org/10.3389/fnbeh.2021.632906>
- Schein AI, Bauer GR (2019) The intersectional discrimination index: development and validation of measures of self-reported enacted and anticipated discrimination for intercategorical analysis. *Soc Sci Med* 226:225–235. <https://doi.org/10.1016/j.socscimed.2018.12.016>
- Schmitt MT, Branscombe NR, Postmes T, Garcia A, Hinshaw SP (2014) The consequences of perceived discrimination for psychological well-being: A meta-analytic review. *Psychol Bull* 140:921–948. <https://doi.org/10.1037/a0035754>
- Shiovitz-Ezra S, Ayalon L (2012) Use of direct versus indirect approaches to measure loneliness in later life. *Res Aging* 34:572–591. <https://doi.org/10.1177/0164027511423258>
- Stadler G, Chesaniuk M, Haering S, Roseman J, Straßburger VM, Schraudner M, Ahmad A-N, Auma M, Banas K, Borde T, Buspavanich P, Dewey M, Di Maio S, Ditlmann R, Enarovic I, Fischer M, Gellert P, Gerstorf D, Grittner U, Gutsell J, Hagelskamp C, Henschel A, Herrmann W, Hilger C, Horozoglu G, Hövener C, Hunter E, Iida M, Keller L, Kendel F, Krumbholz AC, Licha M, Mason K, Mata J, Mau S, Meuwly N, Moschko T, Müller-Werdan U, O'Sullivan JL, Radl J, Rathmann C, Regitz-Zagrosek V, Rieckmann N, Rommel A, Salikutluk Z, Patrick ES, Smith J, Specht J, Stephan P, Stock C, Wenzel P (2023) Diversified innovations in the health sciences: Proposal for a Diversity Minimal Item Set (DiMIS). *Sustain Chem Pharm* 33:101072. <https://doi.org/10.1016/j.scp.2023.101072>
- Stevens C, Liu CH, Chen JA (2018) Racial/ethnic disparities in US college students' experience: discrimination as an impediment to academic performance. *J Am Coll Health* 66:665–673. <https://doi.org/10.1080/07448481.2018.1452745>
- Trigueros R, Padilla A, Aguilar-Parra JM, Mercader I, López-Liria R, Rocamora P (2020) The influence of transformational teacher leadership on academic motivation and resilience, burnout and academic performance. *Int J Environ Res Public Health* 20:7687. <https://doi.org/10.3390/ijerph17207687>

- Wenig V, Heumann E, Stock C, Busse H, Negash S, Pischke CR, Heinrichs K (2023) Associations of loneliness with mental health and with social and physical activity among university students in Germany: results of the COVID-19 German student well-being study (C19 GSWS). *Front Public Health* 11:1284460. <https://doi.org/10.3389/fpubh.2023.1284460>
- Wenig V, Heinrichs K, Heumann E, Lehnchen J, Burian J, Deptolla Z, Stock C (2025) Social-ecological factors associated with loneliness in university students: results from the German cross-sectional StudiBiFra study. *Front Psychiatry*. <https://doi.org/10.3389/fpsyt.2025.1469811>
- Wilson LC, Liss M (2023) Belonging and loneliness as mechanisms in the psychological impact of discrimination among transgender college students. *J LGBT Youth* 20:705–723. <https://doi.org/10.1080/19361653.2022.2049418>
- Zahedi H, Sahebhiagh M H and Sarbakhsh P (2022) The magnitude of loneliness and associated risk factors among university students: A cross-sectional study. *Iran J Psychiatry* 17:411–417. <https://doi.org/10.18502/ijps.v17i4.10690>

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